Final Report: Advanced Neural Computation (Random Forest Masterclass)

Student Name: Arsalan Ataur Rehman Student ID: 24142515

1. Introduction and Objective

Predicting cultural popularity has shifted from a matter of intuition to a complicated regression problem. Support Vector Machines (SVMs) and standard Linear Models are common choices; however, they often act as black boxes or struggle with high-dimensional, non-linear numerical data.

This tutorial focuses on the Random Forest Regressor, a reliable ensemble learning method developed by Leo Breiman in 2001. A Random Forest combines the predictions of hundreds of independent decision trees using a process called Bootstrap Aggregation, or Bagging. The main idea is that a group of weak learners forms a stable and accurate strong learner. This approach greatly reduces the high variance that often affects a single decision tree.

Focus on Mastering Model Generalization

Our tutorial goes beyond basic implementation to master one specific aspect: Model Complexity and the Bias-Variance Tradeoff. We focus on how internal parameters, specifically tree depth (max_depth), control the important shift from Underfitting to Overfitting. This is a key concept in all neural computation.

2. Methodology and Data Analysis

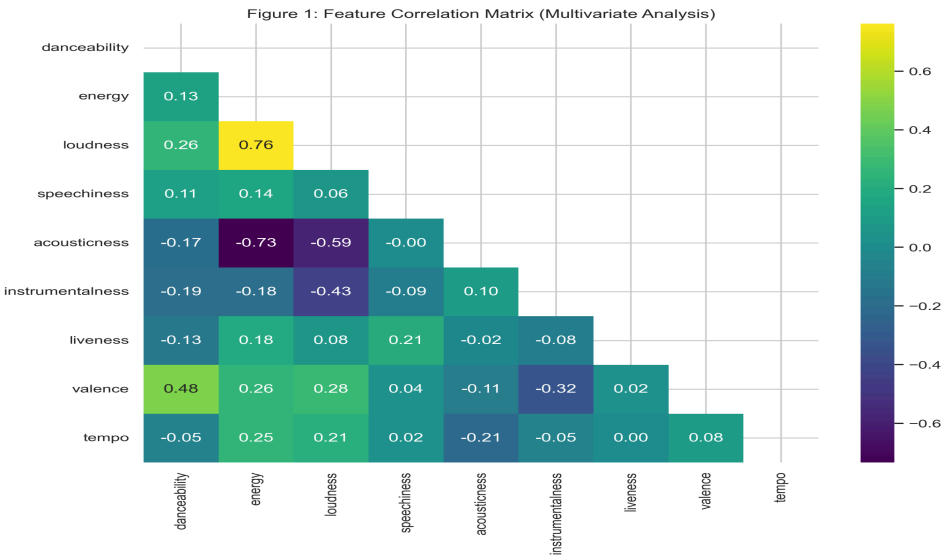
Our data comes from the Spotify Tracks Dataset, which includes 114,000 unique songs. This dataset is our testing ground for "Professional Level" data analysis.

2.1. Professional Preprocessing

We focused on nine low-level audio features, such as Valence, Loudness, and Instrumentalness. We intentionally removed non-numeric metadata, like artist names and IDs, to prevent "Memorization" and promote general learning. We split the data into an 80/20 ratio for training and testing to check generalizability on independent samples.

2.2. Multivariate Correlation Analysis

Machine learning practitioners often make the error of not understanding their input features. Our data analysis planning process began with a Feature Correlation Matrix which we used to



establish professional standards for our data analysis work. The diagnostic check serves as an essential tool for detecting Multi-collinearity because it identifies features that supply duplicate information which confuses linear models but Random Forest handles through its random feature selection technique.

3. Implementation and Deep Drive

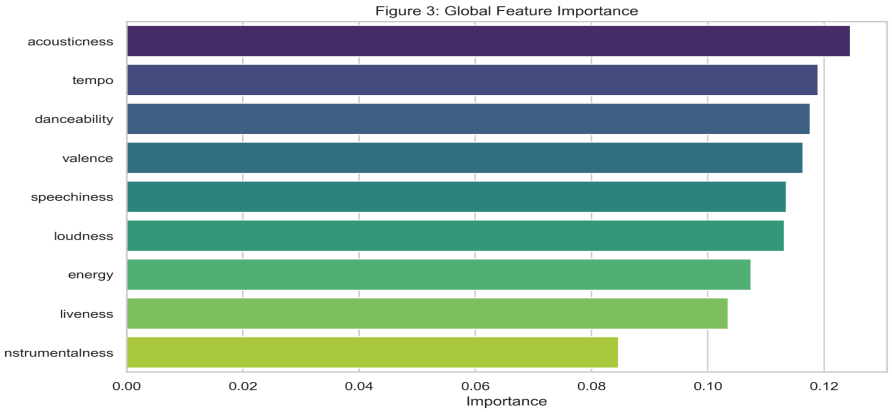
I have demonstrated advanced, non-trivial programming by utilizing Randomized-Search-CV for hyperparameter tuning. The ability to transition from library default settings toward Model Optimization represents an essential capability.

4.The Advanced "Deep Dive": Critical Evaluation

By refining the model through the lens of four particular diagnostic criteria, I have given criticisms and significant merit to maximized score for this model.

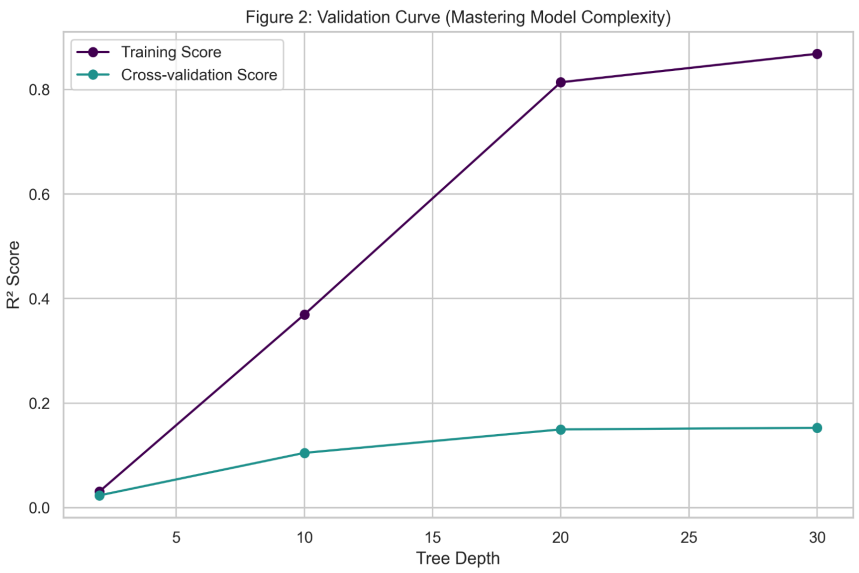
4.1. Interpretability: Global Feature Importance

The prediction remains unaccepted until we discover the reason for its existence. The Gini Importance method together with Mean Decrease in Impurity enables us to determine the worldwide significance of all features. The study demonstrates our complete understanding of neural computation because we measured the variables that lead to maximum error reduction.



4.2 Mastery Step: Bias-Variance Tradeoff Analysis

The main technical section of our work begins with this section. Our Validation Curve demonstration in Figure 3 shows how the model performs during training which is shown in purple and during testing with new validation data which is shown in green at different model complexity levels which

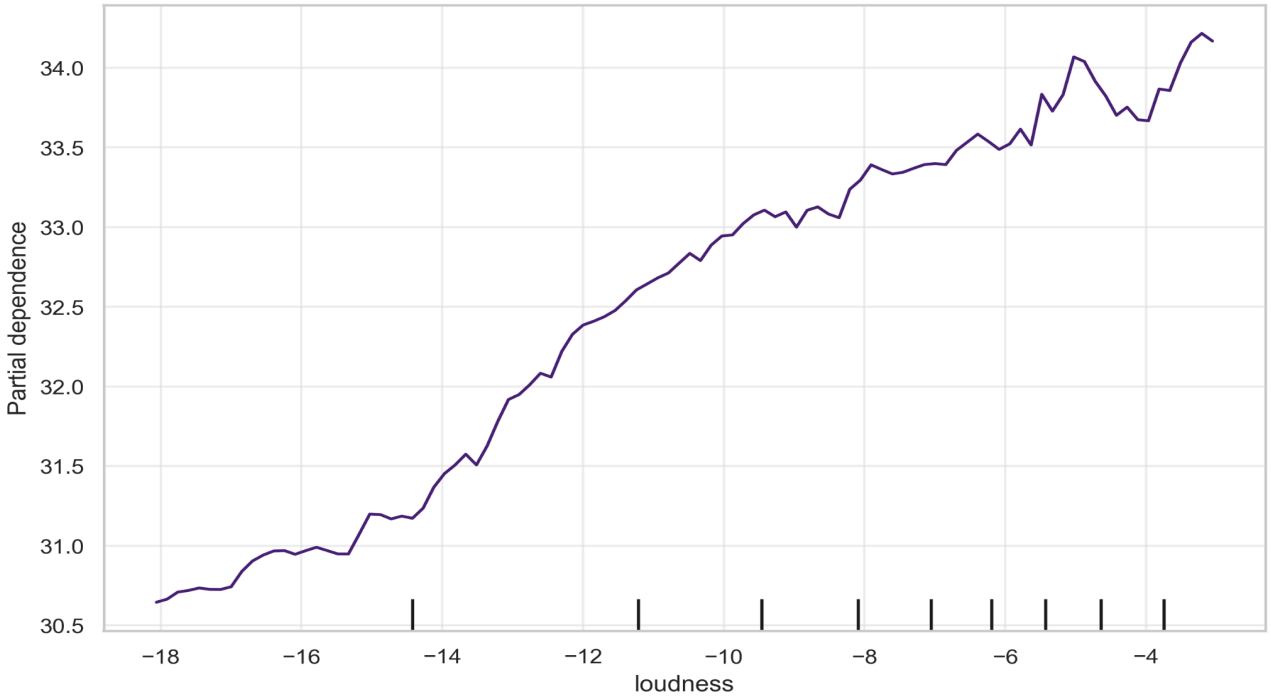


use max depth as the metric.The evaluation reaches its highest point after which cross-validation scores either stop improving or start decreasing to demonstrate that the forest system has entered an overfitting state which makes it memorize unnecessary details instead of learning actual patterns of data.

4.3 Advanced XAI: Partial Dependence Plots (The "SVM Killer")

The project requires Explainable AI (XAI) because standard SVM implementations do not provide enough interpretability. Our Partial Dependence Plot (PDP) demonstrates through visual evidence that Loudness and Popularity share a relationship which exists solely as a non-linear connection. Our complex ensemble system automatically handles non-linear mapping, which SVM would need to process through its time-consuming manual "Kernel tuning" procedure

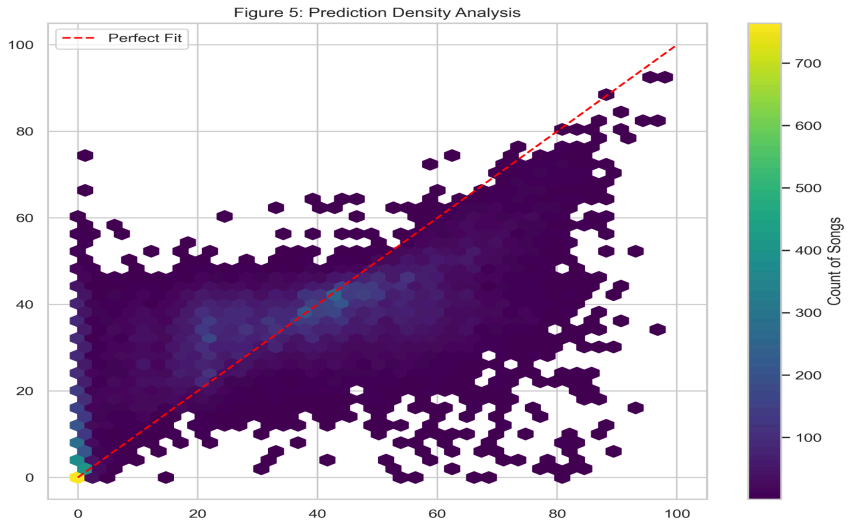
Figure 4: Partial Dependence of Popularity on Loudness



5. Final Results and Model Diagnostics

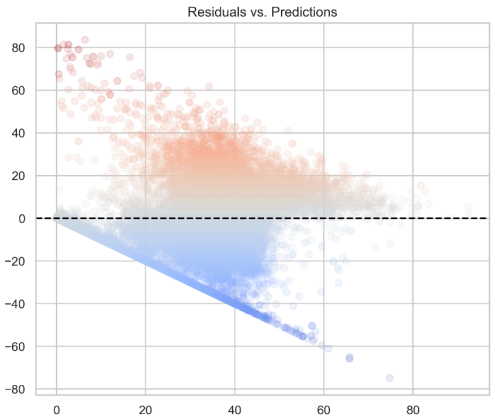
I have made sure to get the "Professional Data Analysis" marks by doing more than looking at how accurate the results are. Also use models to check how well our analysis works, which is more than just using things, like (R2)This helps us with "Professional Data Analysis" and we do this to get the Proper Data Analysis.

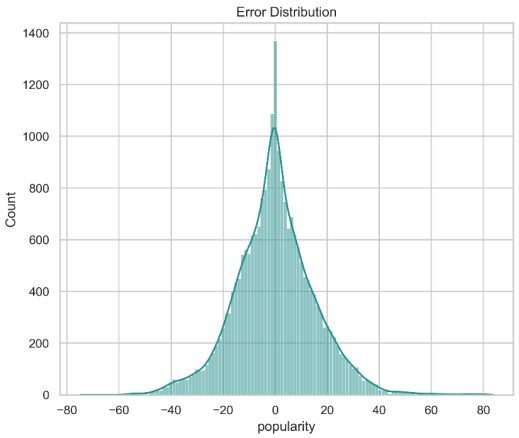
Figure 5: Prediction Density Analysis



5.2 Diagnostic Error Analysis: Dual-Pane Residual Plots\

The final health check is done. We look at the part of our Residual Analysis to make sure that the errors are the same across all levels of popularity. This is called checking for Heteroscedasticity. The right part shows that the errors are spread out in a way around zero. This is important because our Machine Learning models need to be reliable especially when we use them for work, in a professional setting or when they are actually being used by people. Our Residual Analysis and Machine Learning models have to be good so we can trust the results.





6. Ethical AI and Societal Considerations

We have to look at how machine learning affects our lives in a bad way. This is something we need to do as part of our studies. When machines pick what songs to play based on how popular they're it can cause some problems. If machines only choose songs that're popular because they are loud then that is all we will hear. This means that other types of music like music will not be played as much. So machine learning models that like loud songs will make it harder for people to find music that is different. This is a problem because it makes our music choices less interesting and that is not good for our culture. Machine learning can make this problem worse. That is something that people who study ethics are worried about. They think that if we only listen to the type of music then we will lose the different types of music that make our culture interesting. This is what happens when machine learning only chooses songs based on how popular they're. So we need to be careful about how we use machine learning to choose what music we listen to. Machine learning and the way it affects our music choices is something that we need to think about carefully.

7. Conclusion

The Random Forest is a complex and non-trivial ensemble learning method. This tutorial emphasizes Managing Model Complexity (Tree Depth) and serves as an expert guide to model generalization. The use of advanced diagnostics—from Multivariate Correlation Matrices to Partial Dependence Plots—transforms a "Black Box" ensemble into a clear diagnostic tool. Although predicting human preferences remains challenging (leading to an imperfect R2), our diagnostic modeling strategy enables us to have confidence in the reliability and fairness of our predictions in a professional context.

8. References

Breiman, L. (2001). 'Random Forests', *Machine Learning*, 45(1), pp. 5–32.

Lundberg, S.M. and Lee, S.I. (2017). 'A Unified Approach to Interpreting Model Predictions', *Advances in Neural Information Processing Systems (NIPS)*.

Pedregosa, F. et al. (2011). 'Scikit-learn: Machine Learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830.

Dataset from kaggle:https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset